

A good beginner-friendly project is Iris Flower Classification using Python and Scikit-learn.

I can build the complete project including:

1. Dataset & preprocessing
2. Machine Learning model training
3. Model evaluation
4. Prediction program
5. Project folder structure
6. README.md
7. requirements.txt
8. GitHub upload steps
9. Final submission checklist

PROJECT: IRIS FLOWER CLASSIFICATION

Goal: Predict the species of an iris flower from its sepal and petal measurements.

Algorithm: Random Forest Classifier

Library: Scikit-learn

Language: Python

Dataset: Built-in Iris dataset

iris-flower-classification/

```
|
|— data/
|— src/
|   |— train_model.py
|
|— models/
|   |— iris_model.pkl
|
|— README.md
|— requirements.txt
|— .gitignore
```

2.PYTHON CODE — iris_classification.py

```
<\>python
```

Build a complete AI or Machine Learning project using the skills learned.

Publish your project on GitHub with a proper README file.

Review your work and submit the final project through the portal. Solve this question

For your AI/ML internship task, a good beginner-friendly project is Iris Flower Classification using Machine Learning. It demonstrates the full workflow: data loading → preprocessing → model training → evaluation → prediction → GitHub README.

Project: Iris Flower Classification

Objective: Build a machine-learning model that predicts the species of an iris flower from its sepal and petal measurements.

Technologies: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook.

1. Project structure

iris-ml-project/

```
|
|— iris_classification.py
|— requirements.txt
|— README.md
|— .gitignore
```

2. Python code — iris_classification.py

Iris Flower Classification

AI/ML Internship Project

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

1. Load dataset

```
iris = load_iris()
```

```
X = pd.DataFrame(
    iris.data,
    columns=iris.feature_names
)
```

```
y = pd.Series(iris.target, name="species")
```

```
print("Dataset shape:", X.shape)
print("\nFirst 5 rows:")
print(X.head())
```

2. Check for missing values

```
print("\nMissing values:")
print(X.isnull().sum())
```

3. Split data into training and testing sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

4. Feature scaling

```
scaler = StandardScaler()
```

```
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 5. Train Machine Learning model
model = LogisticRegression(max_iter=200)

model.fit(X_train_scaled, y_train)

# 6. Make predictions
y_pred = model.predict(X_test_scaled)

# 7. Evaluate model
accuracy = accuracy_score(y_test, y_pred)

print("\nModel Accuracy:", accuracy)

print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))

# 8. Confusion matrix
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Iris Classification - Confusion Matrix")
plt.tight_layout()
plt.show()

# 9. Test a new flower
new_flower = [[5.1, 3.5, 1.4, 0.2]]

new_flower_scaled = scaler.transform(new_flower)

prediction = model.predict(new_flower_scaled)
```

```
print("\nNew Flower Prediction:")
print("Predicted species:", iris.target_names[prediction])
```

3.requirements.txt

```
pandas
numpy
scikit-learn
matplotlib
seaborn
jupyter
```

4.readme.md

+iris flower classification -README

IRIS FLOWER CLASSIFICATION USING MACHINE LEARNING

PROJECT OVERVIEW

This project uses Machine Learning to classify iris flowers into three different species:

- Setosa
- Versicolor
- Virginica

The model uses four measurements of an iris flower:

- Sepal length
- Sepal width
- Petal length
- Petal width

A Logistic Regression algorithm is trained using the Scikit-learn Iris dataset.

Objective

The main objective of this project is to understand and implement a complete Machine Learning workflow, including:

- Loading a dataset
- Exploring the data
- Checking for missing values
- Splitting data into training and testing sets
- Feature scaling
- Training a Machine Learning model
- Making predictions
- Evaluating model performance
- Visualizing the results

TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib

- Seaborn
- Jupyter Notebook

MACHINE LEARNING ALGORITHM

Logistic Regression

Logistic Regression is a supervised Machine Learning classification algorithm.

It is used in this project to predict the species of an iris flower based on its four numerical features.

DATASET

The project uses the built-in Iris dataset provided by Scikit-learn.

The dataset contains:

- 150 observations
- 4 input features
- 3 Flower species

PROJECT WORKFLOW

Iris Dataset | Data Exploration ↓ Train/Test Split ↓ Feature Scaling ↓ Logistic Regression ↓ Model Prediction ↓ Performance Evaluation ↓ Confusion Matrix
Model Evaluation

The model is evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

The exact accuracy should be taken from the output produced when the program is executed rather than being manually claimed.

HOW TO RUN PROJECT

Step 1: CLONE THE REPOSITORY

<https://github.com/YOUR-USERNAME/iris-ml-project.git>

Step 2: OPEN THE PROJECT FOLDER

```
cd iris-ml-project
```

STEP 3: INSTALL DEPENDENCIES

```
pip install -r requirements.txt
```

Step 4: RUN THE PROGRAM

```
python iris_classification.py
```

EXAMPLE PREDICTION

For the input:

5.1, 3.5, 1.4, 0.2

the model predicts the corresponding iris species.

FUTURE IMPROVEMENTS

The project can be improved by:

- Comparing multiple Machine Learning algorithms
- Adding a graphical user interface
- Creating a Streamlit web application
- Deploying the application online
- Adding cross-validation
- Performing more detailed exploratory data analysis

AUTHOR

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AI&ML Internship project

5. Publish it on GitHub

Create a new GitHub repository named

Iris.ml.project

Then run:

```
<\>bash
```

```
git init
```

```
git add .
```

```
git commit -m "Initial commit - Iris ML Classification Project"
```

```
git branch -M main
```

```
git remote add origin https://github.com/YOUR-USERNAME/iris-ml-project.git
```

```
git push -u origin main
```